MLOPS (S1-25_AIMLCZG523)

Assignment – 1 (Group 85)

MLOps Experimental Learning Assignment: End-to-End ML Model Development, CI/CD, and Production Deployment Experimental Learning

Exploratory Data Analysis and Modelling Choices

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Problem Statement: Build a machine learning classifier to predict the risk of heart disease based on patient health data, and deploy the solution as a cloud-ready, monitored API.

Objective: Design, develop, and deploy a scalable and reproducible machine learning solution utilising modern MLOps best practices. The assignment emphasises practical automation, experiment tracking, CI/CD pipelines, containerization, cloud deployment, and monitoring-mirroring real-world production scenarios.

Exploratory Data Analysis and Modelling Choices

1. Introduction

This document provides a comprehensive analysis of the Exploratory Data Analysis (EDA) and modelling choices made for the Heart Disease UCI classification project. The objective is to build a scalable, production-ready machine learning solution to predict heart disease risk based on patient health metrics, following MLOps best practices.

2. Dataset Overview

2.1 Data Source and Acquisition

- **Dataset**: Heart Disease UCI Dataset
- **Source**: UCI Machine Learning Repository
- **Repository Link**: https://archive.ics.uci.edu/ml/machine-learning-databases/heart-disease/processed.cleveland.data
- **Acquisition Method**: Programmatic download via urllib for reproducibility
- **Format**: CSV with 14 features and 1 binary target variable
- **Total Records**: 303 patient records

2.2 Feature Description

The dataset consists of 13 medical features and 1 target variable:

Feature	Type	Description
age	Continuous	Patient age in years (29–77)
sex	Binary	1 = Male, 0 = Female
cp	Categorical	Chest pain type (1–4)
trestbps	Continuous	Resting blood pressure (94–200 mmHg)
chol	Continuous	Serum cholesterol (126–564 mg/dl)
fbs	Binary	Fasting blood sugar > 120 mg/dl (0 = No, 1 = Yes)
restecg	Categorical	Resting electrocardiographic results (0–2)
thalach	Continuous	Maximum heart rate achieved (71–202 bpm)

exang	Binary	Exercise-induced angina (0 = No, 1 = Yes)
oldpeak	Continuous	ST depression induced by exercise (0.0–6.2 mm)
slope	Categorical	Slope of ST segment (1–3)
ca	Categorical	Number of major vessels (0–3)
thal	Categorical	Thalassemia type (3–7)
target	Binary	Heart disease presence (0 = Absent, 1 = Present)

2.3 Problem Statement

Binary Classification Task: Predict the presence (1) or absence (0) of heart disease based on patient health indicators.

Context: Early detection of heart disease is critical for patient outcomes. An automated classifier can assist healthcare professionals in risk assessment and decision-making.

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3. Exploratory Data Analysis

3.1 Data Cleaning Process

The raw dataset required systematic cleaning to ensure quality for model training:

1. **Missing Value Handling**: Original dataset used "?" to represent missing values. These were converted to NaN and imputed using median values for each feature, preserving the distributional properties of the data.
2. **Data Type Conversion**: All features were converted to float64 for numerical compatibility with scikit-learn pipelines.
3. **Target Encoding**: Original target variable had values 0–4 (representing degrees of heart disease). For binary classification, we converted: values > 0 → 1 (disease present), value = 0 → 0 (no disease).
4. **Validation**: Post-cleaning validation confirmed no remaining missing values and proper data type consistency.

3.2 Dataset Statistics

Post-cleaning dataset statistics reveal important characteristics:

Statistic	Count	Mean	Std Dev	Missing (%)
Total Records	303	—	—	0.0
Age (years)	303	54.4	9.0	0.0
Blood Pressure (mmHg)	303	131.7	17.6	0.0
Cholesterol (mg/dl)	303	246.7	51.8	0.0
Max Heart Rate (bpm)	303	149.6	22.9	0.0
Target Variable (Disease)	303	0.46	0.50	0.0

3.3 Class Distribution Analysis

Target Variable Distribution:

- Class 0 (No Heart Disease): 160 records (52.8%)
- Class 1 (Heart Disease Present): 143 records (47.2%)
- **Class Imbalance Ratio**: 1.12:1 (relatively balanced)

Finding: The dataset exhibits minimal class imbalance, eliminating the need for SMOTE or class weighting strategies. Both classes are well-represented for training.

3.4 Feature Distribution Insights

Continuous Features:

- **Age**: Approximately normal distribution, mean = 54.4 years, range = 29–77 years. Most patients are middle-aged to elderly.
- **Blood Pressure**: Mean resting BP = 131.7 mmHg, indicating moderate hypertension on average. Distribution is relatively symmetric.
- **Cholesterol**: Mean = 246.7 mg/dl (above normal threshold). Wide range (126–564) suggests diverse metabolic profiles in the population.

- **Maximum Heart Rate**: Mean = 149.6 bpm. Normal distribution with slight left skew. Critical indicator of cardiac capacity.
- **ST Depression (oldpeak)**: Right-skewed distribution with median = 0.8 mm. 25% of patients have zero exercise-induced ST depression.

Categorical Features:

- **Sex**: 68% male, 32% female. Reflects historical dataset composition and higher prevalence of recorded heart disease in males.
- **Chest Pain Type (cp)**: Predominantly types 3–4 (typical and atypical angina), representing ~70% of records.
- **Thalassemia (thal)**: Majority fixed (type 3) and reversible (type 7) defects.

3.5 Correlation Analysis

Key Findings from Feature Correlation:

- **Strong Correlations with Target**:
 - **ca (num major vessels)**: r = 0.44 (moderate positive) — More blocked vessels indicate higher disease likelihood.
 - **thal**: r = 0.32 — Thalassemia type moderately associated with disease.
 - **exang (exercise-induced angina)**: r = 0.29 — Strong symptom indicator.
- **Age-related Patterns**: Age shows weak positive correlation with disease (r = 0.24), expected in cardiovascular risk.
- **Low Feature Collinearity**: No feature pairs exceed |r| = 0.70, minimizing multicollinearity concerns. StandardScaler preserves interpretability while normalizing scales.

4. Feature Engineering and Preprocessing Strategy

4.1 Preprocessing Pipeline

A reproducible sklearn ColumnTransformer pipeline was implemented:

- **StandardScaler**: Applied to all 13 features
  - Transforms features to zero mean, unit variance
  - Essential for distance-based and regularized models (Logistic Regression, Random Forest tree splits benefit from normalized scale)

- Preserves information while enabling fair coefficient comparison

4.2 Feature Selection Rationale

Retained All 13 Features because:

- All features carry clinical significance in cardiology
- Correlation analysis showed no severe multicollinearity
- Removing any feature risks losing predictive signal
- Models (especially Random Forest) can handle irrelevant features through internal feature importance ranking

4.3 Train-Test Split Strategy

Approach: Stratified 5-fold cross-validation (not simple hold-out split)
Benefit: Utilizes all data for training while maintaining class balance in each fold
Evaluation: Average metrics across 5 folds reduce variance and provide robust performance estimates

5. Modelling Approach

5.1 Model Selection Rationale

Two baseline classification models were selected to establish performance benchmarks:

5.1.1 Logistic Regression

Why:

- Probabilistic predictions suitable for medical risk stratification
- Interpretable coefficients for feature importance understanding
- Linear decision boundary appropriate for this problem
- Fast training and inference for production deployment

Hyperparameters:

- max_iter = 1000 (convergence iterations)
- Default regularization (L2, C=1.0) to prevent overfitting

Expected Strengths: Interpretability, computational efficiency, probability calibration
Expected Weaknesses: Linear assumption may miss non-linear relationships

5.1.2 Random Forest Classifier

Why:

- Captures non-linear relationships through ensemble of decision trees
- Handles mixed feature types (continuous, categorical) naturally
- Provides feature importance rankings
- Robust to outliers and missing data

Hyperparameters:

- n_estimators = 200 (tree ensemble size)
- random_state = 42 (reproducibility)
- Default max_depth (unlimited) for full tree growth

Expected Strengths: Non-linear pattern capture, feature importance, ensemble robustness
Expected Weaknesses: Less interpretable black-box, higher computational cost, potential overfitting

5.2 Evaluation Metrics

Comprehensive evaluation ensures clinical and operational readiness:

Metric	**Rationale and Interpretation**
Accuracy	Overall correctness: (TP + TN) / Total. Useful baseline but can mislead with imbalanced targets.
Precision	True Positive Rate among predicted positives. Critical in medical context to minimize false alarms (false positives).
Recall (Sensitivity)	True Positive Rate among actual positives. Essential for identifying all disease cases; missing cases are dangerous.

ROC-AUC	Area under receiver operating characteristic curve. Threshold-independent metric evaluating discrimination ability across all probability thresholds. Gold standard for binary classification.

Metric Priority for Model Selection: ROC-AUC chosen as primary metric because:

- Threshold-independent evaluation
- Balances sensitivity (recall) and specificity
- Suitable for imbalanced datasets
- Medical decision-makers can adjust threshold based on business requirements (prioritizing recall vs. precision)

5.3 Model Training and Evaluation Process

1. **Pipeline Construction**: Each model combined with StandardScaler preprocessing into an sklearn Pipeline, ensuring:
   - Scaling fit only on training data (prevents data leakage)
   - Consistent preprocessing during inference
   - Production-ready reproducibility

2. **Cross-Validation**: 5-fold stratified cross-validation evaluates generalization:
   - Trains 5 separate model instances on different subsets
   - Evaluates on held-out folds
   - Averages metrics across folds for robust estimate
3. **Experiment Tracking**: MLflow logs all metrics:
   - Model name and hyperparameters
   - Cross-validation scores (accuracy, precision, recall, ROC-AUC)
   - Model artifacts for reproducibility

5.4 Model Comparison and Selection

Decision Criteria: Best model selected by maximum ROC-AUC score from cross-validation.

Model	**Accuracy**	**Precision**	**Recall**	**ROC-AUC**
Logistic Regression	~0.87	~0.86	~0.88	~0.92
Random Forest	~0.89	~0.88	~0.91	~0.94
Best Model Selected	**Random Forest (Higher ROC-AUC)**			

Rationale for Random Forest Selection:

- Higher ROC-AUC (0.94 vs. 0.92) indicates superior discrimination ability
- Higher recall (0.91 vs. 0.88) reduces false negatives, critical for medical applications
- Acceptable trade-off in interpretability for improved predictive performance
- Ensemble approach provides robustness and stability

5.5 Final Model Training

The selected Random Forest model was retrained on the complete dataset (X, y) to maximize information utilization:

> **Rationale**: After model selection via cross-validation, retraining on full data leverages all available examples without overfitting concerns (architecture and hyperparameters already validated).
>
> **Artifact Storage**: Final pipeline saved as joblib pickle file for deployment:
> - Contains fitted StandardScaler and trained RandomForestClassifier
> - 100% reproducible predictions across environments

- MLflow also logs model artifact for experiment tracking

6. MLOps Implementation Highlights

6.1 Reproducibility and Experiment Tracking

- **MLflow Integration**: All model runs logged with:
  - Model hyperparameters (n_estimators, max_iter)
  - Cross-validation metrics
  - Model artifacts (pickle files)
  - Traceable run IDs for audit
- **Deterministic Execution**: Random seeds (random_state=42) ensure reproducible results across runs
- **Version Control**: All preprocessing logic and model code versioned in GitHub

6.2 Preprocessing Pipeline Portability

- Preprocessing pipeline encapsulated in `preprocess.py` module
- Reusable functions (`load_data`, `clean_data`, `get_preprocessor`) ensure consistency between training and inference
- Standardizer fitted on training data prevents data leakage

7. Data Quality and Assumptions

7.1 Data Quality Summary

- **Completeness**: 100% after median imputation (original dataset had ~10% missing values)
- **Consistency**: All features numeric, uniform range handling via StandardScaler
- **Accuracy**: Data sourced from UCI repository, clinically validated benchmark dataset
- **Timeliness**: Static historical dataset; not streaming or real-time

7.2 Key Assumptions and Limitations

1. **Independent and Identically Distributed (IID)**: Assumes test data drawn from same distribution as training data. Real-world deployment may encounter data drift.

2. **Feature Stationarity**: Model assumes feature distributions and relationships stable over time. Periodic model retraining recommended.
3. **Median Imputation**: Missing values imputed with feature medians, assuming missing-at-random (MAR) mechanism.
4. **Class Definition**: Binary target (disease present/absent) simplifies multiclass problem. Clinically, disease severity spectrum lost.
5. **No Protected Attributes Encoding**: Current model does not explicitly account for fairness across demographic groups (e.g., age, sex bias). Fairness audits recommended.

7.3 Data Drift Monitoring Strategy

For production deployment:

- Monitor feature distributions via statistical tests (Kolmogorov-Smirnov) on incoming predictions
- Track model performance metrics (calibration, discrimination) on recent data
- Trigger retraining pipeline if drift detected exceeding thresholds

8. Conclusion

This EDA and modelling analysis establishes a robust foundation for heart disease prediction:

- **Data Quality**: Comprehensive cleaning and preprocessing ensure input data integrity
- **Feature Engineering**: Systematic approach preserves clinical relevance while enabling scalable pipelines
- **Modelling Choice**: Random Forest selected via evidence-based cross-validation, balancing performance and interpretability
- **MLOps Readiness**: Full pipeline reproducibility, experiment tracking, and containerization enable production deployment
- **Continuous Monitoring**: Proposed data drift monitoring ensures sustained model performance

The resulting model is production-ready and integrated into a complete CI/CD pipeline for automated training, testing, and deployment.

Link to code repository

EDA and Training Notebooks are available in repo:

(https://github.com/2024aa05152-collab/heart-disease-mlops)